

Reactive Statements

Q30

STATE (LEGACY)

INTERVIEW QUESTION

Question. What is the purpose and syntactic structure of the `$:` labeled statement in Svelte 3/4?

ANSWER. In Svelte 3 and 4, the `$:` prefix (which utilizes standard JavaScript label syntax) is used to construct reactive statements and reactive declarations. The compiler automatically analyzes the code following the label, infers the local variables acting as dependencies, and instruments the compiled output to re-execute that specific block of code whenever those dependencies change. A reactive declaration defines a new variable (e.g., `$: x = y * 2`), while a reactive statement runs arbitrary side-effects without variable declaration (e.g., `$: console.log(y)`).

App.svelte

```
<script>
  let count = 0;

  // Reactive declaration: creates a computed variable
  'double'
  $: double = count * 2;

  // Reactive statement: triggers arbitrary side-effects
  $: console.log('Count updated:', count);

  // Reactive block: executes multiple lines upon dependency
  change
  $: {
```

```
if (count > 10) {  
  alert('Threshold reached');  
}  
}  
</script>
```

- ① **Summary.** The `$:` label marks code that should automatically re-run. Svelte handles the dependency tracking statically during compilation, meaning any referenced local reactive variables will automatically trigger the statement when updated.



MAGIC LABEL

**Compiler auto-
tracks
dependencies
here**